

Binary Lifting

“ Concepts & Qns ”



Video-3



codestorywithmik



Cs withMIK



codestorywithMIK

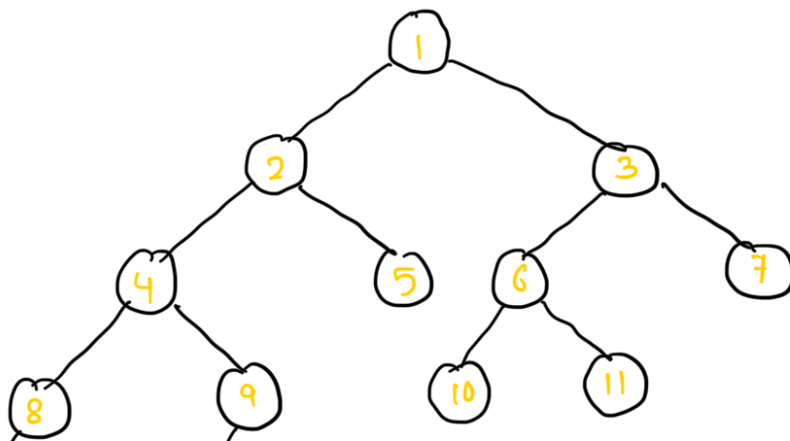
Motivation .



Mehnat Karlo, abhi time hai.
Bad me regret Karne se acha
hai, utilise the time you're right now...

MIK

Finding LCA between two nodes (u, v) using Binary Lifting





(u, v)
 $(12, 13)$

$2^j \leadsto$ Binary lifting.

① Ancestor Table

rows = n

cols = $\log_2 n + 1$

2^0 2^1 $(\log_2 n)$
 $j=0$ $j=1$

node = 0				
node = 1				
node = 2				
⋮				
⋮				

2^j
 $j=0 \dots \log_2 n$

cols = $\log_2 n + 1$
 $\downarrow$

$\log_2(n)$

for (int j = cols - 1; j >= 0; j--) {

$\text{if } (\text{AncestorTable}[u][j] == -1) \text{ continue;}$

$\text{if } (\text{AncestorTable}[u][j] \neq \text{AncestorTable}[v][j]) \{$

$u = \text{AncestorTable}[u][j];$

$v = \text{AncestorTable}[v][j];$

$\}$

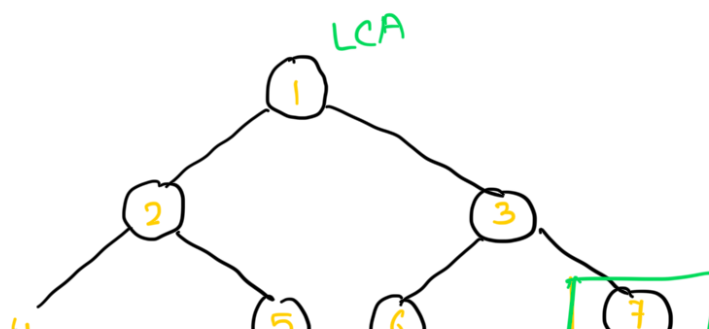
$\}$

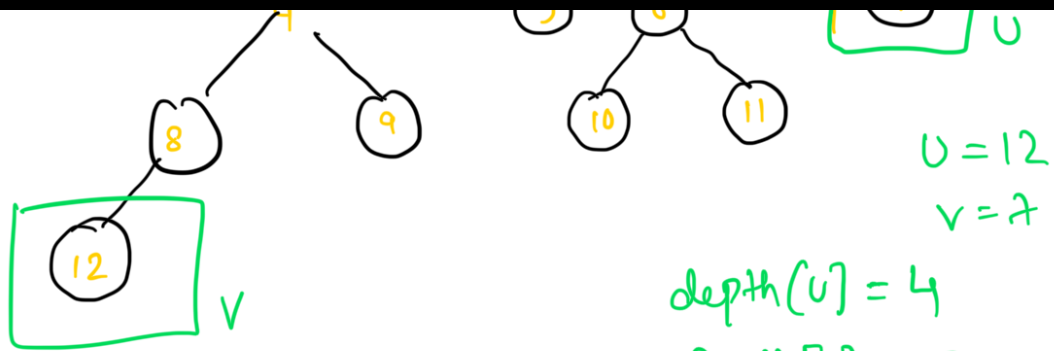
// just below LCA $\rightarrow u, v$

$\text{return AncestorTable}[u][0];$

$\}$

X (u, v) in different depths





$$\text{depth}[u] = 4$$

$$\text{depth}[v] = 2$$

2nd anc of (12)

$$\text{diff} = 4 - 2 = 2$$

Kth ancestor of a node.

$$\text{diff} = K$$

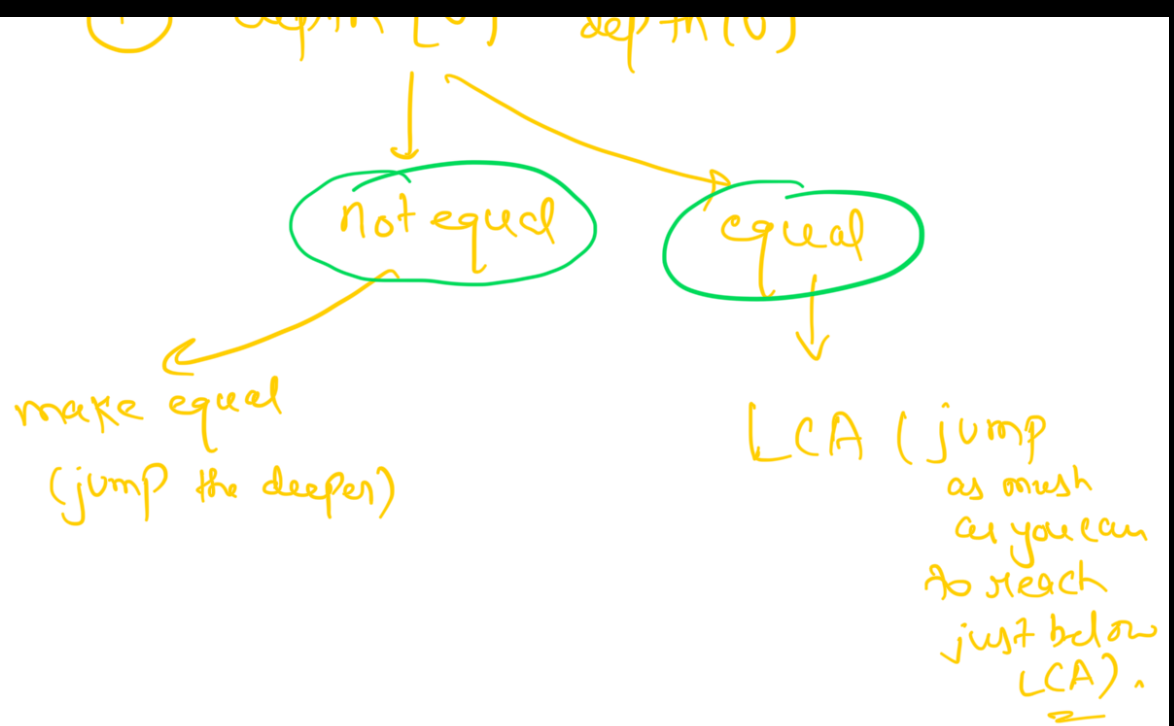
```

if (depth[u] > depth[v])
for (int j=0; j<cols; j++) {
    if (K & (1<<j))
        u = ancestorTable[u][j];
}

```

LCA(u, v)

(i) depth[u] > depth[v]



if ($\text{depth}(u) < \text{depth}(v)$) {

swap (u, v);

}

vide-1,2,3

Copy Paste.

Leetcod → 3559
(POTD).

